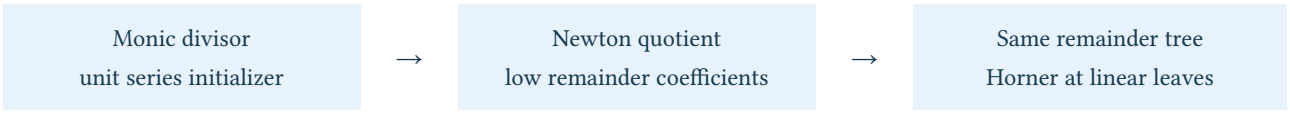
Monic remainder specialization

Prime-field native source · 2026-11-03

The candidate preserves the product/remainder tree, Karatsuba multiplier and Newton reversal. It starts unit-constant series directly, evaluates linear leaves by Horner, and reconstructs only the low remainder coefficients at internal nodes.



84 remainder cases · 10 multipoint cases · 8 complete recoveries

Correctness checked; cost comparison draft validated, not executed

Correctness and construction

If B is monic of degree d, reversal determines Q as before; A minus QB has degree below d, so its coefficients modulo X^d determine the entire remainder. Only low coefficients of Q and B contribute. The internal quotient still exists; the code avoids full-length remainder reconstruction and an unused returned quotient pair. Linear leaves use A mod(X-x)=A(x). Every tree inverse series starts at1, so this evaluation path calls no field-inverse helper.

Four selected native methods pass once: 84 remainder cases, ten multipoint cases against original and Horner references, series/domain errors, and all eight complete public-point recovery receipts. Sixty prior tests are filtered. No candidate cost or timing comparison ran.

Comparison readiness

The canonical protocol validator accepts the baseline/candidate/A-A draft with zero findings. It is not frozen: matched instrumentation/runner, binary identity, preflight, analysis and live EKS admission remain to be completed. The original 4563-case cohort is unchanged.

Evidence and sources

Source locks, reversible routing map, compiler, commands and raw logs accompany receipt.json. See DERIVATION.md for proof and boundaries. The original polynomial implementation and FLINT documentation supply prior context. Protocol validation failures (system jsonschema version and draft enum values) are preserved; the final validation passes.